VI. Transformation of Random Variables (GSE 518)

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Idea: Say $X \sim F_X$. We are usually interested in a function $Y = g(X)$ instead of X itself. What is P_Y, F_Y, f_Y?

Examples:

- Let $X \sim F_X$, with sample space $(-\infty, \infty)$.

$$Y_1 = g_1(X) = X^2$$

$$Y_2 = g_2(X) = \begin{cases} 0 & \text{if } X \leq 0 \\ 1 & \text{if } X > 0 \end{cases} \quad \text{probit and logit models}$$

$$Y_3 = g_3(X) = \begin{cases} 0 & \text{if } X \leq 0 \\ X & \text{if } X > 0 \end{cases} \quad \text{censored models}$$

- X is random $\Rightarrow Y$ is also random.

- $Y = g(X)$ induces a new probability space: $(\mathcal{Y}, \mathcal{B}_Y, P_Y)$

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a) Notes about the Sample Space ($\mathcal{Y}$)

It is convenient to write sample spaces for X and Y as follows:

$$\mathcal{X} = \{x : f_X(x) > 0\} \quad \text{and}$$
$$\mathcal{Y} = \{y : f_Y(y) > 0\} = \{y : y = g(x) \text{ for some } x \in X\}$$

Note: $g(x) : \mathcal{X} \to \mathcal{Y}$, so $\mathcal{X}$ is the *Domain* of $g(X)$ and $\mathcal{Y}$ is the *Range*.

Key: *Always keep track of sample spaces.*

b) Notes about the Borel field or Domain of the Probability Function ($\mathcal{B}_Y$)

It is the same as before, say, the set of all possible subsets of $\mathcal{Y}$ if $\mathcal{Y}$ is countable (i.e., if Y is a discrete random variable), or the set of all intervals in $\mathbb{R}$ if $\mathcal{Y}$ is uncountable (i.e., if Y is a continuous random variable).

Note:

- If X is a discrete random variable, then Y is also discrete random variable.
- If X is continuous random variable then Y can be either continuous or discrete.

c) Notes about the Probability Function (P_Y)

The probability function P_Y depends on P_X. As in prior notes, it is easier to work with *real-valued functions* like the cdf, pmf or pdf than with set functions like P_Y. Hence, we focus on finding the cdf or pmf/pdf of Y when we know the cdf or pmf/pdf of X.

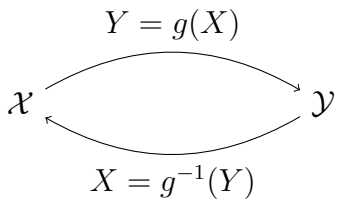
1. Transformation of Random Variables: Initial Ideas

Let $X \sim F_X$, or $X \sim f_X$, and $Y = g(X)$.

Goal: Find F_Y (cdf) or f_Y (pmf/pdf) given F_X or f_X and the fact that $Y = g(X)$).

Notes:

- It is not always easy to find the distribution of Y given the distribution of X and $Y = g(X)$.
- Fortunately, F_Y and f_Y are "easy" to find in some cases, specifically, when $Y = g(X)$ has a *well-defined inverse function* $X = g^{-1}(Y)$, that is, when if $Y = g(X)$ for $x \in X$, then we have that $X = g^{-1}(Y)$ is a *function*.



Examples:

1. $Y = g(X) = \ln(X)$ and $X \in (0, \infty) \Rightarrow X = g^{-1}(Y) = e^Y$, Yes, a function.
2. $Y = g(X) = X^2$ and $X \in \mathbb{R} \Rightarrow X = g^{-1}(Y) = \pm\sqrt{Y}$, not a function.
3. $Y = X^2$ for $X \in [0, \infty) \Rightarrow X = g^{-1}(Y) = +\sqrt{Y}$, is a function.

As mentioned before, note that it is key to keep track of the sample space of X (i.e., of the values that X can take).

1.1. When is $X = g^{-1}(X)$ a (well-defined) function?

Definition: A function $y = g(x)$ is strictly monotone in $\mathcal{X}$ if it is either strictly increasing in $\mathcal{X}$ or strictly decreasing in $\mathcal{X}$ (but not both). We say that:

- A function is strictly increasing in $\mathcal{X}$ if:

$$x_1 > x_2 \Rightarrow g(x_1) > g(x_2), \forall x_1, x_2 \in \mathcal{X}$$
$$\text{or} \quad \frac{dg(x)}{dx} > 0, \forall x \in \mathcal{X}$$

- A function is strictly decreasing in $\mathcal{X}$ if:

$$x_1 > x_2 \Rightarrow g(x_1) < g(x_2), \forall x_1, x_2 \in \mathcal{X}$$
$$\text{or} \quad \frac{dg(x)}{dx} < 0, \forall x \in \mathcal{X}$$

Result 1: If $g(x)$ is strictly monotone in the sample space $\mathcal{X}$, then $g^{-1}(y)$ will be a well-defined function in $\mathcal{Y}$.

Note:

$$g^{-1}(y) = g^{-1}(g(x)) = x$$

2. Finding F_Y and f_Y

2.1. Case 1: When the Random Variable is Continuous

a) How do we find F_Y from F_X (i.e., the cdf of Y given the cdf of X)?

Result 2: (*Theorem 2.1.3 in textbook*)
Let $X \sim F_X$ and $Y = g(X)$.

i) If $g(x)$ is strictly increasing on $\mathcal{X}$:

$$F_Y(y) = F_X\left(g^{-1}(y)\right) \quad \forall y \in \mathcal{Y}$$

ii) If $g(x)$ is strictly decreasing on $\mathcal{X}$:

$$F_Y(y) = 1 - F_X\left(g^{-1}(y)\right) \quad \forall y \in \mathcal{Y}$$

Proof of Strictly Increasing case:

$$F_Y(y) = P(Y \leq y) = P(g(X) \leq y)$$

$$= P\left(g^{-1}(g(X)) \leq g^{-1}(y)\right) = P\left(X \leq g^{-1}(y)\right)$$

$$= F_X\left(g^{-1}(y)\right)$$

(Note that we take inverse of both sides of the inequality and keep the inequality direction the same because $g(x)$ is strictly increasing. The proof of part (ii) of Result 2 is basically the same, except that when the function $g(x)$ is strictly decreasing we have to change the direction of the inequality when taking the inverse of both sides. Due to this, we later use the fact that, for any event A, $P(A) = 1 - P(A^c)$ to write $F_Y(y) = ... = P\left(X \geq g^{-1}(y)\right) = 1 - P\left(X \leq g^{-1}(y)\right) = 1 - F_X\left(g^{-1}(y)\right)$. This is why part (ii) of the result has the part "1−". See example below.)

Example to show how to find $F_Y(y)$ using Result 2:

Let $X \sim F_X$ where $F_X(x) = x$ for $x \in (0, 1)$, $\mathcal{X} = (0, 1)$. This distribution is called a Uniform (0, 1) distribution. Let $Y = -ln(X)$. Find F_Y.

Steps:

1) *Find the sample space of* Y*, that is,* $\mathcal{Y}$ *or (informally) "the values* Y *can take".*

Use $x \in (0, 1)$ and $Y = -\ln(x)$, which implies $\mathcal{Y} = (0, \infty)$.

2) *Check that* $Y = g(X)$ *is strictly monotone on* $\mathcal{X}$.

Given $Y = -\ln(x)$,

$$\frac{dy}{dx} = -\frac{1}{x} < 0 \quad \forall x \in (0,1), \quad \text{thus strictly decreasing.}$$

3) *Get inverse function $x = g^{-1}(y)$. Informally, "solve for x as a function of y."*

$$Y = -\ln(x) \Rightarrow -Y = \ln(x) \Rightarrow \boxed{e^{-Y} = x = g^{-1}(y)}$$

4) *Apply Result 2.*

Applying part ii)

$$F_Y(y) = 1 - F_X(g^{-1}(y)) = 1 - F_X(e^{-y}) = 1 - e^{-y} \quad \text{given } F_X(x) = x$$

Thus,

$$\boxed{F_Y(y) = 1 - e^{-y} \quad \text{for } y \in (0, \infty)}$$

Note: You may also solve the problem above following the same steps as in the proof for part (ii) of Result 2:

$$F_Y(y) = P(Y \leq y) = P(-\ln(X) \leq y) = P(\ln(X) \geq -y) = P(X \geq e^{-y}) = 1 - P(X \leq e^{-y})$$

(Note: The last line uses the fact that, for any event A, $P(A) = 1 - P(A^c)$ along with the fact that X is continuous, so we do not care about writing "$<$" or "$\leq$".)

So, $F_Y(y) = 1 - P(X \leq e^{-y}) = 1 - F_X(e^{-y}) = 1 - e^{-y}$, which is the same result we got above.

b) How do we find f_Y with f_X (i.e., the pdf of Y given the pdf of X)?

Result 3: (*Theorem 2.1.5 in textbook*)

Let $X \sim f_X(x)$, $Y = g(X)$ and assume $g(x)$ is strictly monotone on $\mathcal{X}$. Further, suppose $f_X(x)$ is continuous on $\mathcal{X}$ and $g^{-1}(y)$ has a continuous derivative on $\mathcal{Y}$. Then,

$$f_Y(y) = \begin{cases} f_X(g^{-1}(y)) \cdot \left| \frac{dg^{-1}(y)}{dy} \right| & \text{for } y \in \mathcal{Y} \\ 0 & \text{otherwise} \end{cases}$$

This theorem/formula is usually called the *change of variable method or formula*.

Proof

Recall that the pdf equals the derivative of the cdf (for continuous case), so:

$$f_Y(y) = \frac{dF_Y(y)}{dy}$$

Let $g(x)$ be strictly increasing on $\mathcal{X}$ (the strictly decreasing case is analogous). Then, using Result 2 above to find the cdf of Y ($F_Y(y)$) from the cdf of X ($F_X(x)$) and then taking the derivative to find the pdf, we get (remember to use the Chain Rule!):

$$f_Y(y) = \frac{dF_Y(y)}{dy} = \frac{dF_X(g^{-1}(y))}{dy} = f_X(g^{-1}(y)) \cdot \frac{dg^{-1}(y)}{dy}$$

Example to show how to find pdf $f_Y(y)$ using Result 3:

Continuation of example above. Let $X \sim$ Uniform $(0, 1)$, which implies that $X \in (0, 1)$, $f_X(x) = 1$, and $F_X(x) = x$. Let $Y = -\ln(X)$. Find $f_Y(y)$.

Steps:

1) *Find the sample space of Y, that is, $\mathcal{Y}$ or (informally) "the values Y can take".*

In the previous example about Result 2, we found: $\mathcal{Y} = (0, \infty)$

2) *Check that $Y = g(X)$ is strictly monotone on $\mathcal{X}$.*

We already showed in the example about Result 2 that the function is strictly decreasing on $\mathcal{X}$.

3) *Get inverse function $x = g^{-1}(y)$ and its derivative with respect to y.*

In the example about Result 2 we found the inverse function to be:

$$x = g^{-1}(y) = e^{-y}$$

Taking its derivative with respect to y we find:

$$\frac{dx}{dy} = -e^{-y}$$

4) *Apply Result 3.*

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right|$$

$$= (1) \cdot | - e^{-y}| = e^{-y}$$

$$\Rightarrow \boxed{f_Y(y) = \begin{cases} e^{-y} & \text{for } y \in (0, \infty) \\ 0 & \text{otherwise} \end{cases}}$$

(Note the importance of (i) specifying the sample space of $\mathcal{Y}$ and (ii) taking the absolute value when the function is strictly decreasing, as otherwise we would get a *negative* pdf, which is clearly wrong.

2.2. Case 2: When the Random Variable is Discrete

a) How do we find f_Y from f_X (i.e., the pmf of Y given the pmf of X)?

Result 4:

Let X be a discrete random variable with pmf $f_X(x)$ on $\mathcal{X}$, and let $Y = g(X)$, where $g(X)$ is strictly monotone on $\mathcal{X}$ with inverse function $x = g^{-1}(y)$. Then, Y is discrete with pmf given by:

$$f_Y(y) = \begin{cases} f_X(g^{-1}(y)) & \text{if } y \in \mathcal{Y} \\ 0 & \text{otherwise} \end{cases}$$

- Note: there is no term $\left|\frac{dg^{-1}(y)}{dy}\right|$ in the discrete case.

Example to show how to find pmf $f_Y(y)$ using Result 4:

Let $\mathcal{X} = \{0, 1\}$, $f_X(1) = P(X = 1) = p$, $f_X(0) = P(X = 0) = 1 - p$ for $0 < p < 1$, or

$$f_X(x) = \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases}$$

or (all are equivalent),

$$f_X(x) = p^x(1 - p)^{1-x}$$

This is called a *Bernoulli* distribution, and we write $X \sim Bernoulli(p)$.

Let $Y = 1 - X$. Find $f_Y(y)$.

Steps:

1) *Find the sample space of Y, that is, $\mathcal{Y}$ or (informally) "the values Y can take".*

Given $\mathcal{X} = \{0, 1\}$ and $Y = g(X) = 1 - X$, we have that $\mathcal{Y} = \{0, 1\}$.

2) *Check if $g(X)$ is strictly monotone on $\mathcal{X}$.*

$$Y = 1 - X \Rightarrow \frac{dY}{dX} = -1 < 0 \Rightarrow \text{yes, it is strictly decreasing on } \mathcal{X}$$

3) *Find inverse function $x = g^{-1}(y)$. Informally, "solve for x as a function of y."*

$$Y = 1 - X \Rightarrow X = g^{-1}(Y) = 1 - Y$$

4) *Apply Result 4.*

$$f_Y(y) = f_X(g^{-1}(y)) = f_X(1 - y) = p^{1-y}(1 - p)^y$$

$$\boxed{f_Y(y) = \begin{cases} (1-p)^y p^{1-y} & \text{if } y \in \{0, 1\} \\ 0 & \text{otherwise} \end{cases}}$$

Hence, Y follows a Bernoulli distribution with probability $(1-p)$, or $Y \sim \mathit{Bernoulli}(1-p)$.

3. Final Notes

Why is it easier to get F_Y or f_Y when $Y = g(X)$ is strictly monotone and $X = g^{-1}(Y)$ is a (well-defined) function?

- Otherwise we have to take care of all possible values of X that give a specific value $y \in \mathcal{Y}$.

- For example, let $Y = X^2$ for $X \in \mathbb{R}$. Now, for example, the value $y = 4$ can come from $X = 2$ *or* $X = -2$, so

$$P(Y = 4) = P(X = 2) + P(X = -2)$$

If our function is not monotone, one option is to (i) break the space $\mathcal{X}$ into pieces that are strictly monotone; (ii) apply Result 3 (if pdf) or Result 4 (if pmf); and (iii) add the resulting "pieces" to get the final pdf/pmf.